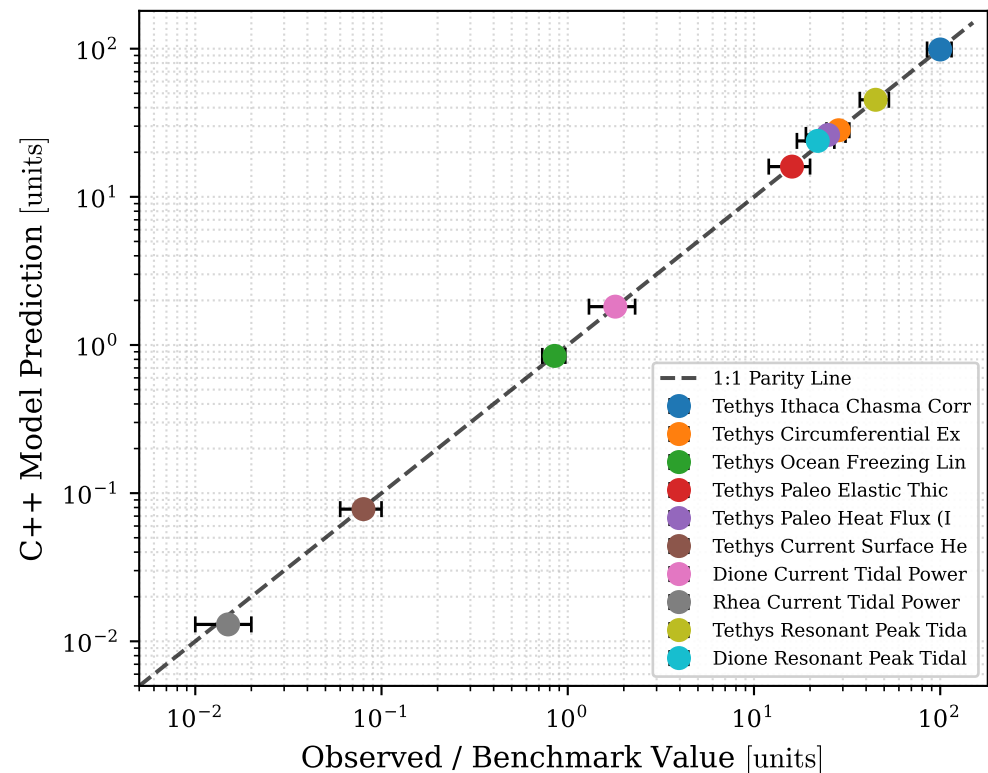
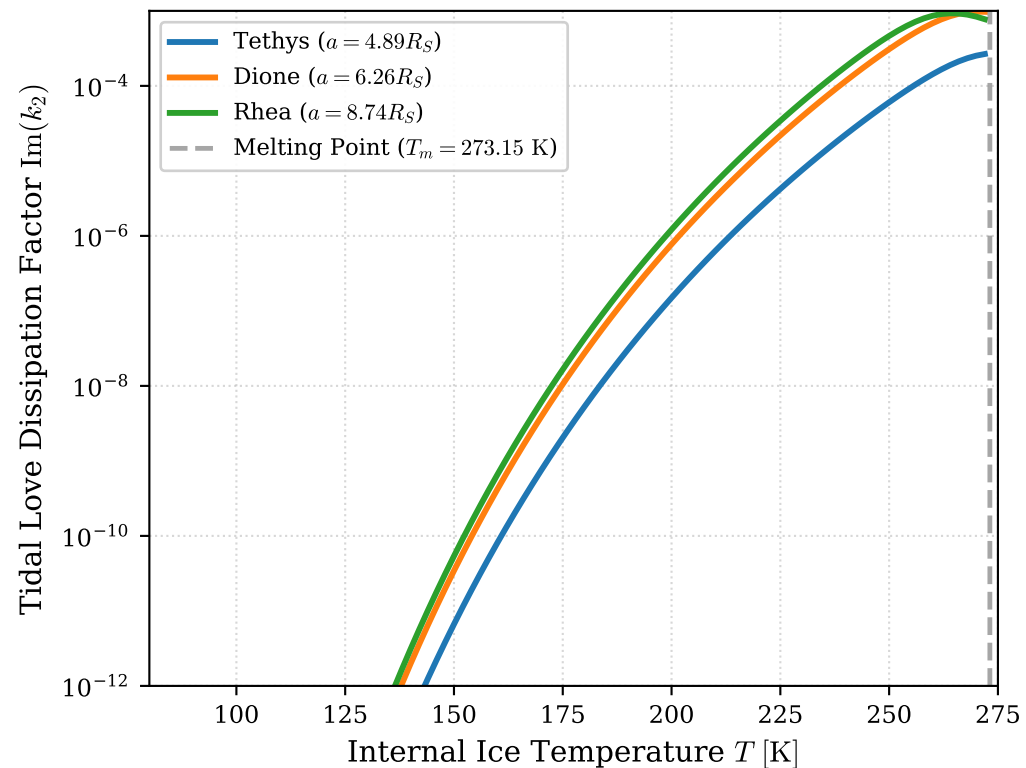


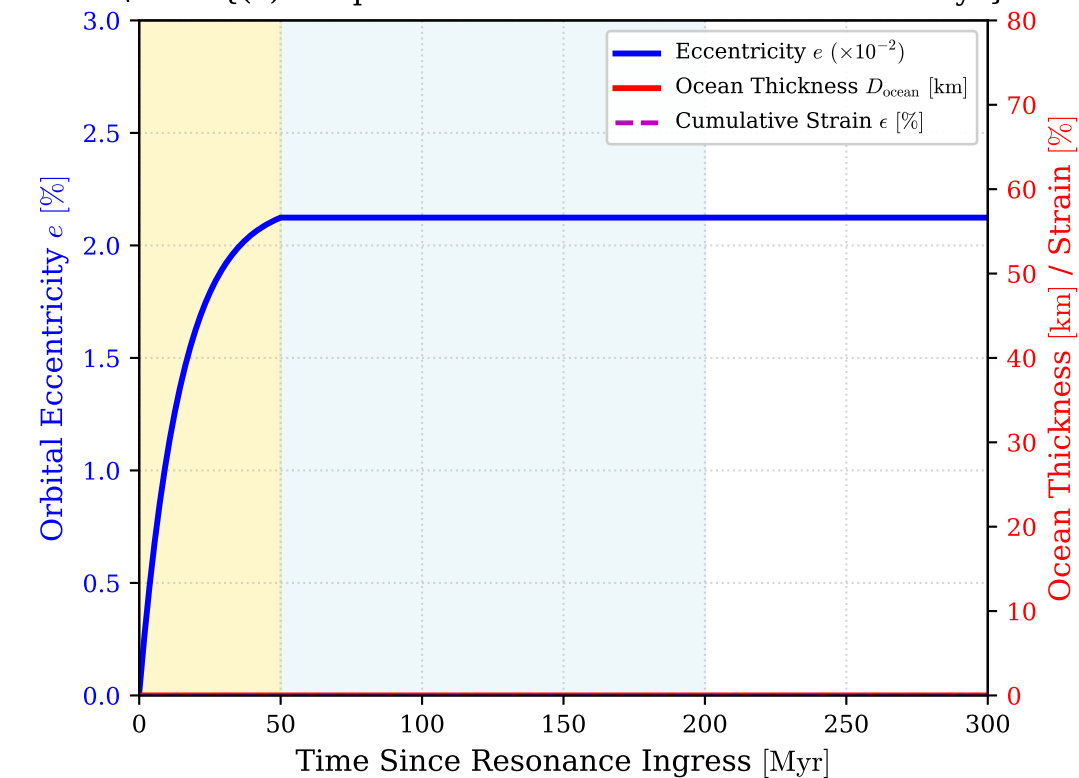
(a) Observational Benchmark Validation ($R^2 = 0.9992$)



(b) Viscoelastic Tidal Dissipation Response



(c) Coupled Thermal-Orbital Evolution of Tethys



(d) Ocean Freezing & Ithaca Chasma Geometry

